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THOUGHT TO TEXT CONVERSION USING DEEP LEARNING

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ABSTRACT: This paper intends to design a very simplified model of the human brain that learns tasks by mimicking the behavior of brain patterns. An EEG (Electroencephalography) signal is a physiological approach to record the electrical signal pertaining to the brain activity by placing the iron affinity electrode i.e. metal disc sensors on the scalp. The magnitude of this signal is quite small, measured in microvolts. There are five types of waves, namely- Alpha, Beta, Theta, Gamma and Delta. This EEG signal is acquired by placing the electrode at the frontal and temporal lobe positions which are then sampled and amplified using Original Muse. MATLAB or Python (open-source libraries) will be used for the processing of the raw EEG signal. This raw EEG signal denoting a band of thought impulse is fed to deep learning framework by threshold sampling. The network will be trained by supervised learning algorithm and simultaneously made to learn robust high-level feature from the processed EEG signal. Further testing of the neural network with a few predefined words under numerous test conditions will be done.

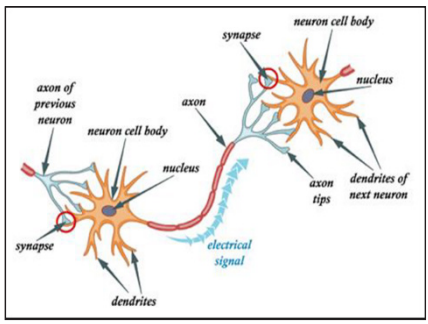
Keywords: EEG, Muse, MATLAB, Deep Learning, Electroencephalography

I. INTRODUCTION

Billions of cells in our brain called neurons process electric signals to interpret information. Dendrites provide external stimuli for the neuron, processed in the neuron cell body. This signal is progressed through the Axon to the subsequent neuron. Depending on the potency of the signal it is chosen to either abide or renounce it. Every neuron can interconnect with 200,000 other neurons over 100 different classes on neurons. Electric impulses passed between synapses and dendrites causes the release of specific chemical transmitter substance from the sending side junction. This results in a fluctuation of electric potential inside the destination cell. If this potential equals a certain threshold, then the cell is excited and a pulse/action potential of fixed energy and duration is sent out to axons of other cells.

Artificial Neural Network is a valuable information processing method which mirrors the characteristics of the biological neural network of the human brain. This network contains a large number of densely interconnected processing parts: nodes or neurons.

These neurons are connected to having an interconnect weight (which is a measure of the "importance" of the input). The aggregate behaviour is portrayed by the ability to learn, evoke and stereotype training patterns or data performing identical to the human brain.

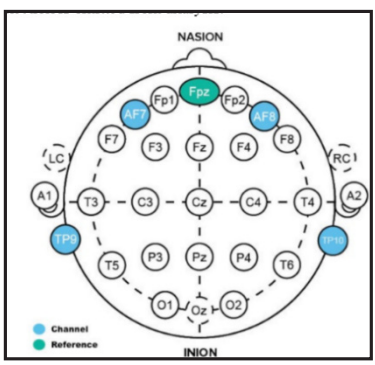


(Source: https://www.deeplearningtrack.com/single-post/2017/07/09/Introduction-to-NEURAL-NETWORKS-Advantages-and-Applications)

Figure 54.1 Neuron.

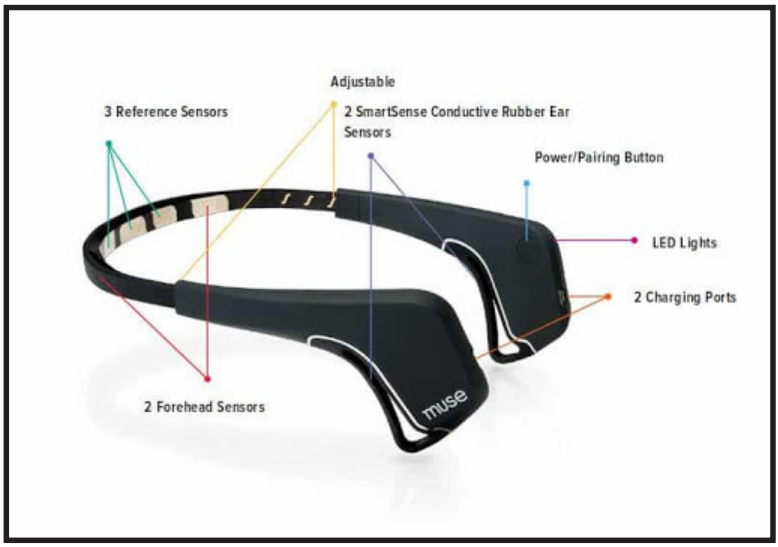
II. LITERATURE SURVEY

Many EEG signal acquiring devices are available in the market with various types, MUSE headset (Figure 54.1.) was chosen as this device has 4 electrodes with a sampling rate of 256Hz and also less computational time is required for acquiring EEG signal and it is much cost-efficient than other EEG headband. [Olave E. Krigolson] Muse headband also has Bluetooth and by default notch filter of frequency, 60Hz is set.



(*Source: https://isl.co/work/zone/)*

Figure 54.2 Muse headband.



(Source: https://www.researchgate.net/publication/308497256)

Figure 54.3 Electrode location.

Muse headband with four sensors that is TP9, AF7, AF8, TP10 (Figure 54.2) with the data set extracted of each mental state tested a combinations of features classifier models and selection algorithm to define their performance and recognition accuracy and number of features more needed, with different features and classifiers used to get an accuracy close to 87%. [Jordan J. Bird] Muselsl is used to convert the acquired EEG signals from MUSE to Micro Volts and record the data into a primary dataset for extraction of features.

EEGLAB is a toolbox or graphical user interface were further signal processing is been done. A location of files with channel in different formats including Matlab, binary, Biosemi EDF, ASCII, EGI, European standard, Snapmaster and Neuroscan. From the above formats, Matlab array and EDF format can be easily used across the various cross-platform processing. [Arnaud Delorme] Available functions such as Channel, EEG data, data visualization preprocessing which has artifact rejection, Independent Component Analysis (ICA), epoch selection, averaging, Filtering and time/frequency decompositions including channel based on resampling of data.

EEG signals can be classified by means of feature extraction from EEG signals. Set of features can be classified as Auto-regression, Linear Complexity, Energy Entropy and Energy Spectrum Density. [Nandish] Neural networks play an important role in increasing performance and feature extraction further extracting features on Max & Min and Average method of the data set can be determined.

To classify EEG signal using Deep neural network to get the relationship between the corresponding brain activities and EEG data. Various classification of EEG signals can be seen, Layer such as Autoencoder is used to remove artifacts such as background noise, [Xiang Zhang] Conventional recurrent neural network (CRNN) learns high-level feature presentation through dense low embedding from EEG signal acquired.

Muse headband supports low energy Bluetooth so for streaming and acquiring the data MUSE LSL or MUSE lab stream or MUSE monitor was used to get the Raw EEG Signal (Alpha, Beta, Delta, Theta) After which different feature algorithm and classifier will be used to check the performance. A positive event-related potential (ERP) P300 that occurs around 300ms after getting an unexpected stimulus. P300 may be a higher-level cognitive processing, for example, conscious attention. [John Griffiths] The N170 happens around 170ms after face perception and T5 and T6 are the lateral electrodes. A large negative event-related potential (ERP) component N170 that happens after the detection of faces, but not on objects, scrambled faces, or other body parts.

Frequency Band Name	Frequency Bandwidth	State Associated with Bandwidth	Example of Filtered Bandwidth
Raw EEG	0–45 Hz	Awake	
Delta	0.5–3.5 Hz	Deep Sleep	
Theta	4–7.5 Hz	Drowsy	
Alpha	8–12 Hz	Relaxed	
Beta	13–35 Hz	Engaged	

(Source: https://ieeexplore.ieee.org/document/7975182)

Figure 54.4 EEG Spectrum Classification.

III. IMPLEMENTATION

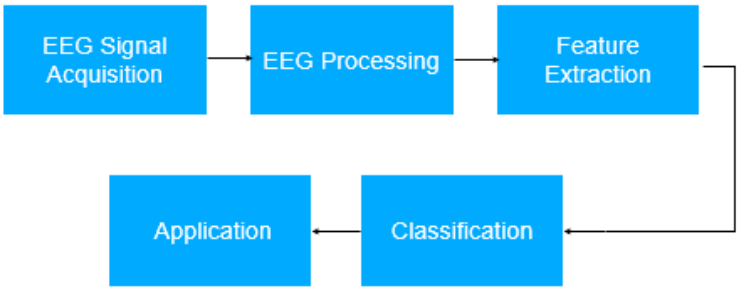


Figure 54.5 Block diagram of the system.

A. EEG Signal Acquisition

The real-time estimation of operator brain status into the computer interface holds a vast encouragement for the feature extraction of the thoughts from the EEG signal. The Muse headband was connected to the machine interface using Bluetooth low energy (BLE) operating at a standard spectrum range of 2.400–2.4835 GHz ISM band. Muselsl is a simple all-in-one approach to live streaming experiment data between applications in a lab. This fundamental transport library is compatible with various language interfaces such as C, C++, Python, Java, and MATLAB. Stream outlets, stream inlets, and built-in-clock are the abstractions of liblsl library used with python language. The streaming outlet sends a single - multichannel data sample by sample into the lab network with uniform value types (integers, floats, doubles, strings) at regular or irregular sampling rates. Once, the streaming is initialized the Stream Inlets records the EEG signal for 60 seconds at the sampling rate of 256 Hz and stores the signal from 4 channel electrodes and 1 reference channel with a timestamp into csv format. In an alternative method, muse monitor (an android based 3rd party brainwave recording application) developed by James Clutterbuck was used to stream EEG signal. The streaming data was recorded for 30 seconds with the vertical scale of 100 uV and stored in csv format. The EEG signal streaming and recording is done in a quiet environment. By default, both the muselsl and muse monitor uses a notch filter of 50 Hz for India standard power.

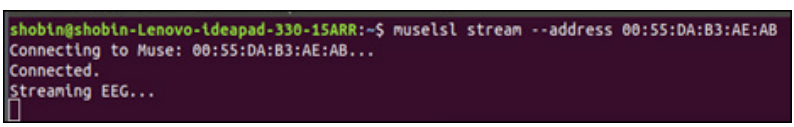


Figure 54.6 Connecting Muse headband and start the streaming layer using MUSELSL.

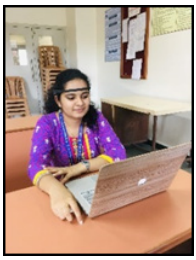
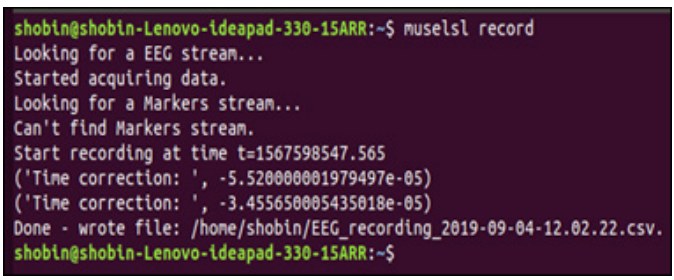


Figure 54.7 Student research participant with Muse headband in a classroom.



```
shobin@shobin-Lenovo-ideapad-330-15ARR:~$ muselsl record
Looking for a EEG stream...
Started acquiring data.
Looking for a Markers stream...
Can't find Markers stream.
Start recording at time t=1567598547.565
('Time correction: ', -5.520000001979497e-05)
('Time correction: ', -3.455650005435018e-05)
Done - wrote file: /home/shobin/EEG_recording_2019-09-04-12.02.22.csv.
shobin@shobin-Lenovo-ideapad-330-15ARR:~$
```

Figure 54.8 Recording the EEG Signal for 60 seconds.

B. EEG Signal Processing

The acquired EEG signal was processed using EEGLAB an open-source toolbox which runs in MATLAB environment. EEGLAB includes functions such importing information about data and channel, visualising data and pre-processing the EEG signal which includes artifact rejection, filtering, epoch extraction and Independent Component Analysis (ICA).

EEGLAB has following functions: Importing data, channel information, and event location. It includes various formats like binary, EGI, MATLAB, Biosemi EDF, CSV, ASCII and BDF, Data epoching and data filtering, Baseline Removal and Artifact Rejection.

1. *Baseline Removal*

It is a method which is used to remove a mean value from each epoch in a data where the differences are high between consecutive data epochs which are unnecessary and can affect the data. The baseline period can be specified in ms to remove the mean in the dataset. It is done by clicking Remove Baseline option in tools.

2. *Filtering*

EEG signals are very low voltage signals and different types of noises alter the signals. To reduce the effect of common noise differential amplifiers are used that can be present on the electrodes. The differential amplifier has one input as a single electrode

and other is the reference electrode. All these amplifiers are low-noise amplifiers with high gains. A capacitor is used to remove DC component in the signal. The notch filter is used to remove the selected frequency band. It does not alter other frequencies which belong to the EEG signal. To reduce the effect of the voltage from the galvanic skin response across the head, the EEG may use a filter to attenuate frequencies below 5 Hz. A low pass filter is needed to reject noise at higher frequencies than the EEG signals. The filtering was accomplished by using filters functions. It is necessary to filter out the high-frequency component from raw EEG data. Using passband filter with lower cut-off frequency as 0.1Hz and higher cutoff frequency as 10 Hz.

$$3.1\, H\left(e^{j\omega}\right) = \begin{cases} e^{j\omega} & \begin{cases} \text{for} -\omega_c - \omega_b < \omega < -\omega_c + \omega_b \\ \text{for } \omega_c - \omega_b < \omega < \omega_c + \omega_b \end{cases} \\ 0 & \text{Otherwise} \end{cases}$$

3. *Artifact Removal*

While collecting EEG signals, artifacts are very important which can affect the data. Artifacts can deteriorate the quality of EEG data. Artifacts are undesired signals which arise from error while experimenting, eye blink and external noise.

EEGLAB has a tool to apply and evaluate the outcome of independent component analysis (ICA) of EEG data. Independent Component Analysis is a method to segregate independent sources which are linearly mixed from different sensors. The main idea of ICA is to reduce the collective information among projections of data or increase their joint entropy.

For example, while recording EEG on the scalp, ICA helps to segregate artifacts embedded in the data. ICA is used to separate linearly mixed sources. EEGLAB contains various ICA algorithm like runica, binica, jader and sobi. Runica algorithm was used for artifact rejection.

The components xi of the acquired random vector $x = (x_1, \ldots, x_m)^T$ are produced as an addition of the independent components S_k, k = 1,…, n

$$3.2\, x = \sum_{k=1}^{n} S_k a_k$$

where the acquired random vector, x is represented by the basis vectors as given below

$$3.3\, a_k = (a_{1,k,\ldots,} a_{m,k})^T$$

C. Feature Extraction

A feature represents an identifying belonging, an identifiable measuring, and a useful element received from the area of a structure. Selection property is meant to decrease the loss of essential content enclosed in the signal. A collection of different ways on how to decrease these losses has been broadly used in order to extract their features from EEG signals, there are several various methods and some of them are stated

here like eigenvector methods (EM), Fast Fourier transform (FFT), time-frequency distributions (TFD), autoregressive method (ARM), wavelet transform (WT) and so on. These various methods are related to each other having their own advantages and disadvantages which were obstinate.

1. *Fast Fourier Transform (FFT)*

This method does analysis of the EEG data signal using mathematical tools, whose equation is mentioned below. Let $x_0,\ldots, x_{N-1}$ are the series of complex numbers. From the series, the discrete Fourier Transform (DFT) is given by the equation as

$$3.4\ X_k = \sum_{n=0}^{N-1} x_n e^{-i2\pi k/N} \quad k = 0,\ldots,N-1$$

Where $e^{-i2\pi k/N}$ is the original Nth root of 1

2. *Wavelet Transform (WT) Method*

Wavelet transform or WT provides a chief role in the field relating to diagnostician and acknowledgement: It is used to suppress the biomedical signals which are the observations of animals and other organisms to extract information relating to physiological activities of that can vary with time and which also contains many points called as data points which are basically the data collected from the information which are then translated into a minor little framework presenting the signal. We know that the EEG signal is a differential equation with time deviating coefficients hence the only promising mode to extract these features of data points from the raw data i.e, a biomedical signal is by using the time-frequency domain methods like WT (wavelet transform). In this method, the continuous wavelet transforms i.e CWT is a non-numerical (i.e., formal) transform that can provide a complete depiction of the EEG signal by allowing scale parameters to keep on changing along with translating the wavelets of the signal. The function x(t) at the scale of (a>0) $a \in \mathbb{R}^{+*}$ and the translational rate $b \in \mathbb{R}$ is given as below

$$3.5\, X_w(a,b) = \frac{1}{|a|^{1/2}} \int_{-\infty}^{\infty} x(t)\overline{\psi}\left(\frac{t-b}{a}\right) dt$$

Where $\psi(t)$ is a continuous function in both time domain and frequency domain and they can also be called as the mother wavelet.

To regain the original signal x(t), the first reverse continuous wavelet transform can be made use of.

$$3.6\, x(t) = C_\psi^{-1} \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} X_w(a,b) \frac{1}{|a|^{1/2}} \tilde{\psi}\left(\frac{t-b}{a}\right) db \frac{da}{a^2}$$

Where $\tilde{\psi}(t)$ is the dual function of $\psi(t)$ and

$$3.7\, C_{\psi} = \int_{-\infty}^{\infty} \frac{\overline{\hat{\psi}}(\omega)\hat{\tilde{\psi}}(\omega)}{|\omega|} d\omega$$

The constant C is also known as the wavelet admissible constant. The wavelet in which the admissible constant i.e. C satisfies $0 < C_{\psi} < \infty$ is called an admissible wavelet.

D. Classification

Machine learning categorization are used in this study which are now briefly described below.

1. *Support Vector Machine (SVM)*

It is a methodical learning of theory-based classification method. For a given two-class linearly separable classification problem, SVM tries to discover a hyperplane which categorizes the input space with an extreme margin. The ideal hyperplane is created as below

$$3.8\ \ w.x_i + b \geq +1 \text{if } y_i = +1$$

$$3.9\ \ w.x_i + b \leq +1 \text{if } y_i = -1$$

where xi is the ith input vector ($x \in R$), and y_i is the class label of the ith input ($y \in \{-1, +1\}$), w is the weight vector normal to the hyperplane, and b is called the bias. Optimal hyperplane can be found out by using two margins that are parallel.

Margins can be found by the equation below

$$3.10\ \ w.x_i + b \leq \pm 1$$

The input vectors that decide the margins are called support vectors. If the issue or problem is not geometrically separable, then we use a kernel relation that is placed in the input vectors so that the struggle will turn linearly divisible in an altered position

$$3.11\ \ K(x_i, x_j) = \varphi(x)\varphi(x_j)$$

The Effect of the two-class linearly non-separable problem is as shown below

$$3.12\ \ f(x) = \text{sign}\left(\sum_i a_i y_j \varphi(x)\varphi(x_i) + b\right)$$

2. *Multilayer Perceptron (MLP)*

MLP is a non-linear neural network-based method comprising of three sequential layers: input, hidden and output, where the hidden layer transmits the input data to the output layer.

3. *Naïve Bayes (NB)*

The NB classifier provides a simple and efficient probabilistic classification based on bayes' theorem, which proposes that the extracted features are not dependent.

4. *K- Nearest Neighbor (K-NN)*

The k-nearest neighbour is a supervised learning algorithm that identifies a challenging sample's class involving to the utmost of the class of k-nearest training samples.

5. *K-fold Cross Validation*

All classification models in the current work were trained and observed with EEG data and then accepted employing k-fold cross-validation, which is an expected used technique that differentiates among performance of two classification algorithms, or calculates the demonstration of a single classifier on a given dataset.

VI. RESULT

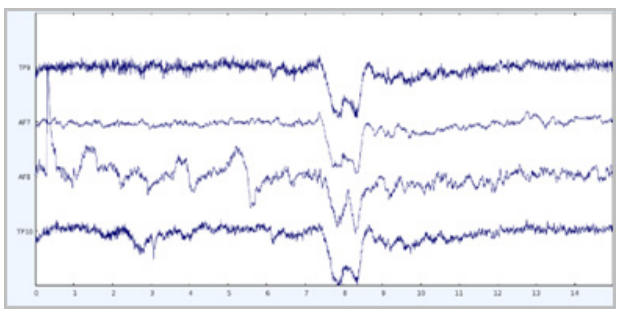


Figure 54.9 Raw EEG Signal (where TP9, TP10, AF7, AF8 are the electrode sensors).

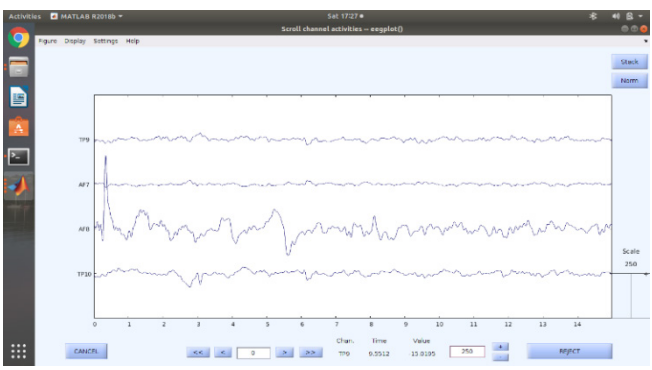


Figure 54.10 Bandpass filtered and baseline corrected EEG signal.

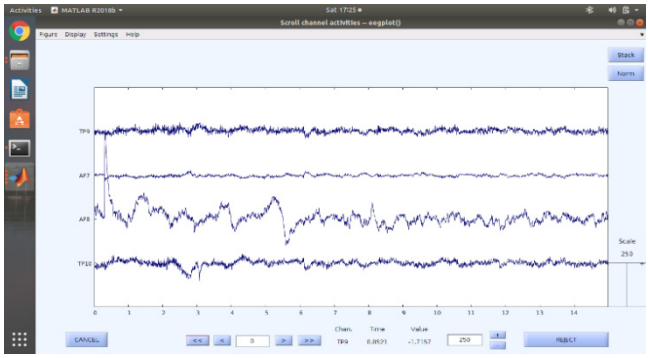


Figure 54.11 Independent component analysis (ICA) pruned EEG signal.

VII. CONCLUSION

In this paper, deep learning model is designed to interpret the raw EEG signals for the purpose of converting the user's thought to text. The proposed approach adopts an ICA to remove artifacts and various feature extraction methods to classify the given signal. Real-world dataset of EEG signal was collected and will be evaluated using our proposed approach. The accuracy of the model can be increased by using various statistical transforms. In future we propose to build a full system whose application will be action gaming with simpler action control for moving forward, backwards, left, right and few more actions like fire, enable/disable scope; communication console will include some emergency words required for immediate attention and help for specially aided people; and various thought based electronic devices. Sooner or later this means of hands-free communication would be used with all measure computer interferences for texting, navigating, or browsing across the various applications. This model improves the efficiency of speech interfaces with devices and it can be useful for those with speech pathologies.

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